



الجامعة المصرية اليابانية للعلوم والتكنولوجيا

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Final Project

MTE 324: Automatic Control 1

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Submitted to

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Abstract

This project presents a LabVIEW-based control system for a telescopic linear actuator driven by a DC motor and guided by position feedback using a reed switch. The actuator uses a gear-driven mechanical system to convert rotational motion into linear displacement. A limit switch acts as a safety cutoff at the maximum extension. The reed switch generates pulses—each corresponding to approximately 1.04 mm (based on 100 pulses per 52 mm)—enabling precise position tracking. The user inputs the desired displacement (ranging from -10 to +10 inches), and the system actuates accordingly. This setup showcases how simple components like the L298 driver, reed switches, and LabVIEW can be integrated into a functional closed-loop actuator system.

Introduction

Linear actuators are essential in various engineering systems where accurate linear motion is required. These actuators find widespread use in manufacturing, robotics, and automation. This project involves designing a telescopic linear actuator system that translates rotational motion from a DC motor into linear displacement. The motion is regulated through a LabVIEW control interface, based on real-time feedback from a reed switch sensor.

System Analysis

Mechanical and Electrical Components:

- Motor: A DC motor rated up to 36V, currently limited to 12V input due to L298N driver constraints.
- Driver: L298N H-Bridge motor driver, chosen for its bidirectional control and ease of use with LabVIEW-Arduino communication.
- Camshaft: Engages a physical limit switch when reaching a designated maximum extension.
- Reed Switch: Provides digital feedback for every (half) rotation, enabling pulse-counting control.
- Arduino UNO: Handles signal acquisition from switches and communicates with LabVIEW.



Fig1. Arduino UNO



Fig2. Motor driver



Fig3. Linear actuator with read

Mathematical Model of the System

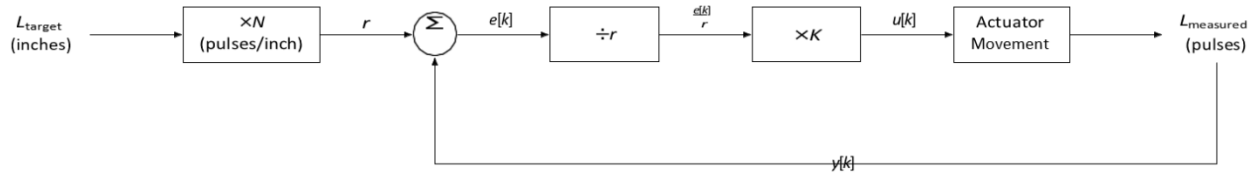


Fig4. Block diagram

Signal Processing Explanation (Step-by-Step):

- The target length input from the user is given in inches as L_{target} .
- The system converts this to pulses using:

$$r = L_{\text{target}} \times N$$

where $N = 50$ pulses/inch (based on the reed switch spec).

- The actuator's current position is measured in pulses and denoted by $y[k]$.
- The error is computed as the difference between the reference and current position:

$$e[k] = r - y[k]$$

- This error is normalized by dividing it over the reference:

$$\frac{e[k]}{r}$$

which ensures scale-independent behavior.

- The normalized error is multiplied by a constant gain K to compute the control input:

$$u[k] = K \cdot \frac{e[k]}{r}$$

- This signal $u[k]$ is sent to the actuator. Based on the sign and magnitude of $u[k]$, the actuator moves forward or backward at each sampling step.
- The system repeats this loop every $T_s = 20$ ms until the measured length L_{measured} reaches the target.

Design Rationale

- Safety: Hardware-based limit switch disables motion at full extension.
- Feedback Control: Digital pulse counting allows real-time estimation of actuator position.
- User Interaction: LabVIEW GUI accepts input in inches, automatically converted to pulses (50 pulses/inch).
- Direction Control: Direction is inferred from sign of displacement (+ or -).

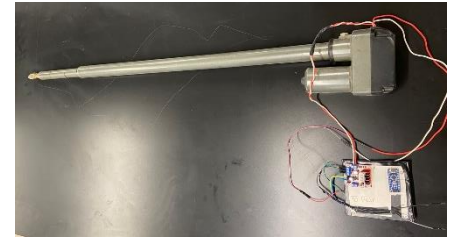


Fig5. Hardware setup

Simulation Insights

LabVIEW simulation validates system behavior and user control logic. Below are example pseudocode snippets used in simulation:

- **Direction Control Logic:**

```
if desired_Positions < pulse_count:
    set_motor_direction(REVERSE)
else:
    set_motor_direction(FORWARD)
```

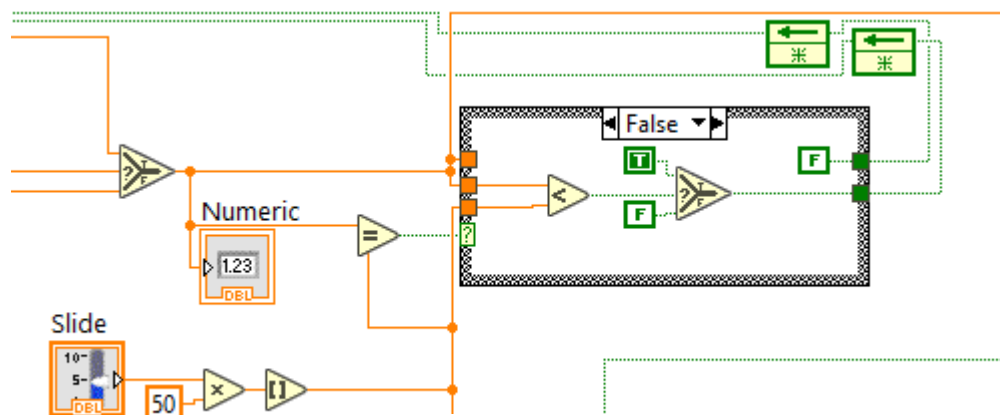


Fig6. Direction Control Logic

- **PWM Value:**

$$P_{wm} = \frac{||Current Position| - |Desired Position||}{|Desired Postion|} * k$$

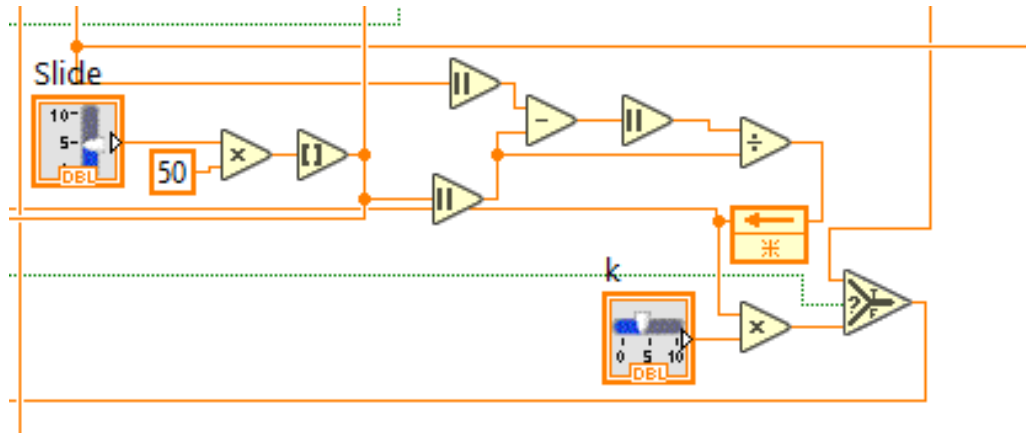


Fig7. PWM Value

- Full Code Diagram:

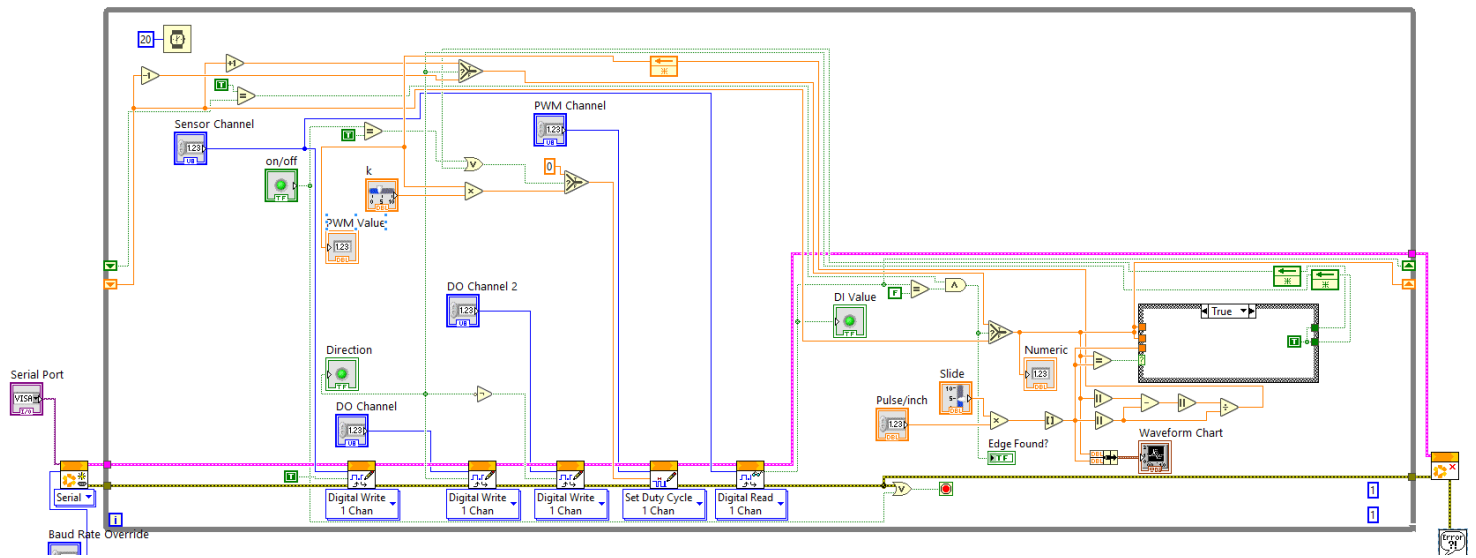


Fig8. Full Code Diagram

- **Front Panel:**

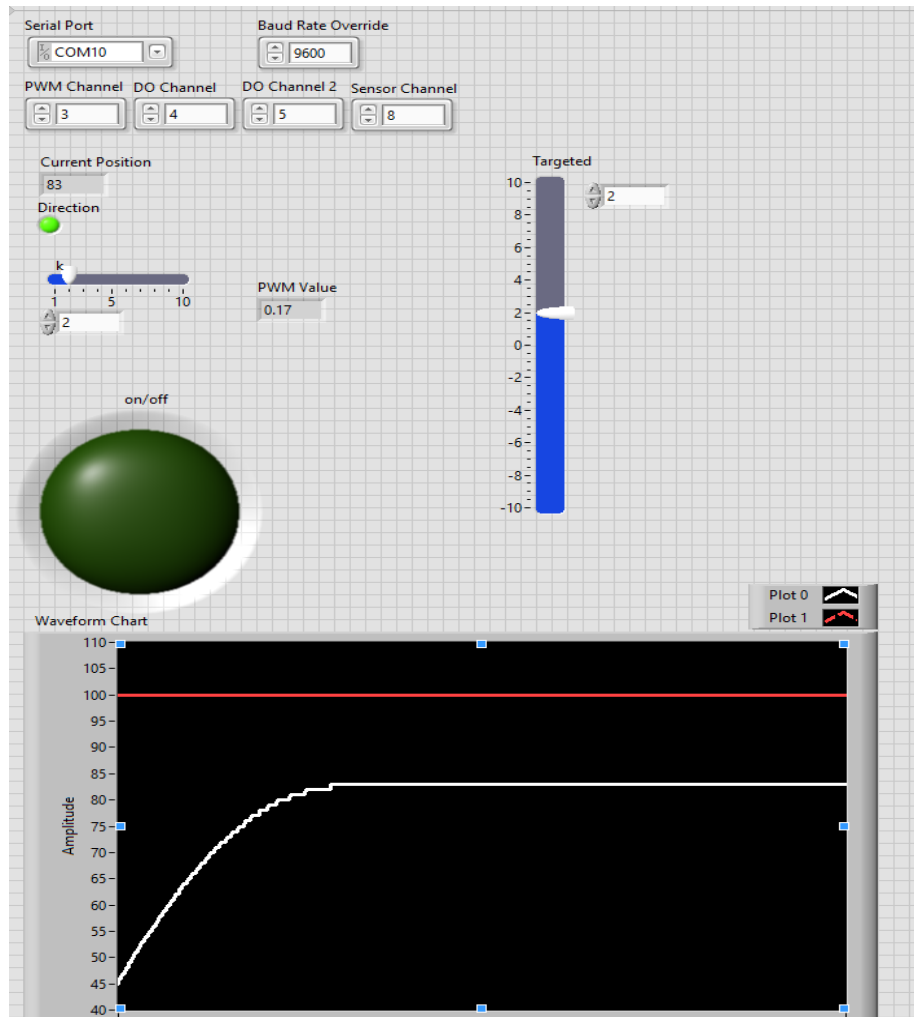


Fig9. Front panel

Applications and Future Improvements

Applications:

- Moving Solar Panels.
- Directing Satellite Dish
- Robotic arms and actuated platforms
- Automated inspection and adjustment systems
- Lifting or positioning systems with predefined travel limits

Future Improvements:

- Upgrade Motor Driver to 36V
- Replace the reed switch with the encoders
- Implement PID closed-loop control
- Integrate wireless communication for remote control

Conclusion

This project successfully demonstrates the control of a linear actuator using a DC motor and feedback from a reed switch. Despite hardware limitations, the system achieves accurate and safe movement control within a ± 10 inch range. LabVIEW enhances usability and facilitates future extensions. This work illustrates the viability of integrating simple sensors and control logic into practical motion control systems.